



# Forex Analysis Dashboard

## COMPREHENSIVE TECHNICAL ANALYSIS PLATFORM

**Overview:** A sophisticated forex trading analysis platform providing real-time market insights, technical indicators, and risk assessment tools. Built on R Shiny with MySQL integration, this dashboard delivers professional-grade analytical capabilities for currency pair trading across multiple timeframes and analytical dimensions.



### Database Connection & Data Management

- MySQL database integration with real-time connectivity
- Configurable connection parameters (host, port, credentials)
- Multi-currency pair data loading and synchronization
- Bid/Ask price filtering with quality validation
- Automated data refresh and update mechanisms
- Connection status monitoring and error handling



### Market Overview

- Real-time price displays with current rates and daily changes
- Interactive candlestick charts with customizable timeframes
- Volume analysis with trading activity visualization
- Price trend identification and pattern recognition
- Key metrics dashboard (open, high, low, close, volume)
- Multi-pair comparison views and heat maps



### Price Analysis & Chart Tools

- OHLC (Open-High-Low-Close) detailed visualizations
- Bid-Ask spread analysis with real-time tracking
- Price distribution histograms and statistical summaries
- Support and resistance level identification
- Time-series decomposition and trend analysis
- Customizable chart intervals (1min, 5min, 1H, 1D, 1W)
- Zoom and pan functionality for detailed examination
- Price forecasting with confidence intervals



### Technical Indicators Suite

#### Momentum Indicators

- ✓ RSI (Relative Strength Index) - 14, 21, 28 periods
- ✓ MACD (Moving Average Convergence Divergence)
- ✓ Stochastic Oscillator with %K and %D lines
- ✓ Momentum indicator with customizable periods

#### Trend Indicators

- ✓ Simple Moving Averages (SMA) - 20, 50, 100, 200
- ✓ Exponential Moving Averages (EMA)
- ✓ Bollinger Bands with standard deviation controls
- ✓ Average Directional Index (ADX)

- Customizable indicator parameters and periods
- Overbought/oversold zone highlighting (RSI 70/30)
- Signal line crossover detection and alerts
- Multi-indicator overlay capabilities

## Volatility Analysis

- ▶ Historical volatility calculations (realized volatility)
- ▶ ATR (Average True Range) for volatility measurement
- ▶ Rolling window volatility with adjustable periods
- ▶ Implied volatility estimations and comparisons
- ▶ Volatility clustering identification
- ▶ Volatility cones and forecasting models

## Risk Metrics & Portfolio Analysis

### Risk Measures

- ✓ Value at Risk (VaR) - 95% and 99% confidence
- ✓ Maximum Drawdown calculation and visualization
- ✓ Sharpe Ratio for risk-adjusted returns
- ✓ Sortino Ratio focusing on downside risk

### Performance Metrics

- ✓ Cumulative returns tracking
- ✓ Win/loss ratio analysis
- ✓ Profit factor calculations
- ✓ Recovery period assessment

## Correlation & Multi-Pair Analysis

- ▶ Dynamic correlation matrix for multiple currency pairs
- ▶ Heatmap visualizations with color-coded strength indicators
- ▶ Statistical significance testing (p-values, confidence intervals)
- ▶ Correlation breakdown analysis (positive/negative trends)
- ▶ Rolling correlation analysis with adjustable windows
- ▶ Correlation network graphs showing pair relationships
- ▶ Time-varying correlation tracking
- ▶ Pearson and Spearman correlation methods

## Key Platform Capabilities

- **Interactive Visualizations:** Plotly-powered charts with zoom, pan, and hover details
- **Real-Time Updates:** Live data synchronization from MySQL database
- **Export Functionality:** Download charts as PNG/SVG and data as CSV
- **Multi-Timeframe Analysis:** Analyze across minutes, hours, days, and weeks simultaneously
- **Customizable Dashboards:** User-defined layouts and indicator combinations
- **Professional Design:** Teal gradient theme with intuitive navigation

**Technology Stack:** R Shiny Framework | Plotly Interactive Charts | MySQL Database | TTR Technical Indicators | DT Data Tables | Quantmod Financial Analysis | RMySQL Connector